



Rossmoyne Senior High School

Semester One Examination, 2021

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One: Calculator-free

SOLUTIONS

WA student number: In figures

M.	9.						
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In words

Your name

Time allowed for this section

Reading time before commencing work: five minutes
Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	52	35
Section Two: Calculator-assumed	13	13	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. NOTE The word “respectively” means in the same order as given earlier.

Section One: Calculator-free

35% (52 Marks)

This section has **eight** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(5 marks)

- (a) Describe the type (using row, column, square, zero or identity) of each of the following matrices:

(i) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

Solution
Square
Specific behaviours
✓ correct type

(1 mark)

(ii) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Solution
Column
Specific behaviours
✓ correct type

(1 mark)

(iii) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Solution
Zero
Specific behaviours
✓ correct type

(1 mark)

- (b) State the size of the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ in the form $m \times n$.

(1 mark)

Solution
3×2
Specific behaviours
✓ correct size

- (c) Write the 4×4 identity matrix I .

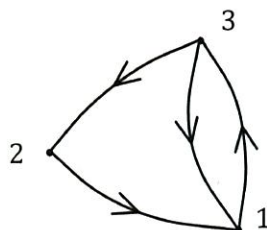
(1 mark)

Solution
$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Specific behaviours
✓ correct matrix

Question 2

(5 marks)

Three people who share a computer network have each forgotten their own password but can remember at least one other person's password. In the following diagram, the directed lines indicate that the person at the start of the line remembers the password of the person at the end of the line. For example, person 2 remembers the password of person 1.



- (a) Create a square matrix M where the entry $M_{i,j}$ is 0 if person i does not remember the password of person j , and is 1 if they do. (3 marks)

Solution	
$M =$	$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$
Specific behaviours	
✓ 3×3 matrix with just zeros and ones ✓ leading diagonal all zeros ✓ correct matrix	

- (b) Determine M^2 . (2 marks)

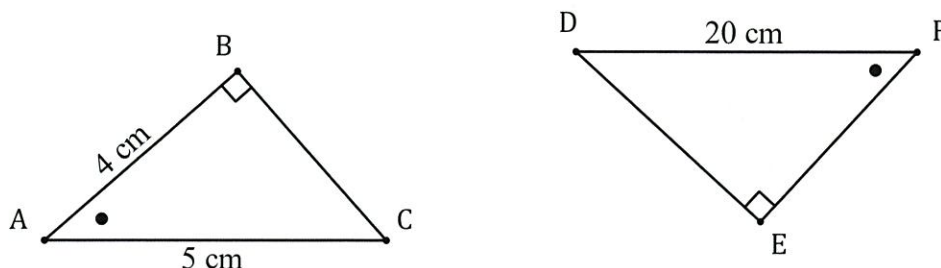
Solution	
$M^2 =$	$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$
Or fills matrix using two-step knowledge of password.	
$M^2 =$	$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$
Specific behaviours	
✓ viable attempt to multiply M by itself ✓ correct matrix	

[ANS ONLY ← CORRECT ✓
 INCORRECT ← ONE WRONG ENTRY ✓x
 TWO OR MORE WRONG xx]

Question 3

(5 marks)

The diagram below, not drawn to scale, shows two similar right triangles.



- (a) Calculate the scale factor for the larger triangle relative to the smaller. (1 mark)

Solution
$20 \div 5 = 4$
The scale factor is 4.
Specific behaviours
✓ correct scale factor

- (b) Determine the length of side BC , the length of side DE and the length of side EF . (3 marks)

Solution
$BC = \sqrt{5^2 - 4^2} = 3 \text{ cm}$
$DE = 3 \times 4 = 12 \text{ cm}$
$EF = 4 \times 4 = 16 \text{ cm}$
Specific behaviours
✓ calculates AC
✓ uses scale factor to calculate DE
✓ uses scale factor to calculate DF

- (c) Calculate how many times greater the area of the large triangle is compared to the area of the small triangle. (1 mark)

Solution
$4^2 = 16$
The area is 16 times larger.
Specific behaviours
✓ squares linear scale factor

f/t

Aus only OK if clearly must have used scale factor from (a).

Accept the square of the scale factor from (a).

Question 4

(7 marks)

A young person has drawn up a budget for their weekly income of \$800. Some of their income is allocated to rent, living expenses, loan repayments, phone, and the rest to savings.

	A	B
1	Weekly budget	
2	Rent	\$235.00
3	Debt repayments	
4	Living expenses	\$290.00
5	Phone	\$65.00
6	Savings	
7	Total	\$800.00

The young person allocates 15% of their weekly income to debt repayments.

- (a) Determine the budget allocation for debt repayments.

(2 marks)

Solution	
$10\% = 80 \rightarrow 5\% = 40 \rightarrow 15\% = 120$	
Allocation is \$120 per week.	
Specific behaviours	
✓ indicates use of an appropriate method ✓ correct allocation	

Ans only < Correct ✓
Incorrect xx

- (b) The young person has been saving for a holiday and needs to save another \$720 to make the booking. If the above budget is maintained, determine the number of weeks will it take them to save this amount.

(3 marks)

Solution	
$235 + 120 + 290 + 65 = 710$	
Weekly savings: $800 - 710 = \$90$	
Number of weeks will be $\frac{720}{90} = \frac{72}{9} = 8$.	
Will take 8 weeks to save the amount.	
Specific behaviours	
✓ indicates weekly savings ✓ indicates correct method for time to save ✓ correct number of weeks	

f/e
f/e

units (weeks) req'd
Word statement
not required.

- (c) If the budget allocations for rent and phone increase by \$7 and \$3 respectively, what effect will this have on your answer to part (b)?

(2 marks)

Solution	
Will save $7 + 3 = \$10$ less, so \$80 per week.	
Number of weeks will be $720 \div 8 = 9$ weeks.	
Will change the time to 9 weeks (or increase the time by 1 week).	
Specific behaviours	
✓ indicates correct savings figure ✓ correctly describes effect on previous answer	

f/e from (b)

Question 5

(5 marks)

- (a) Calculate the value of $p + q \times r$ when $p = 4$, $q = 6$ and $r = 2$.

(1 mark)

Solution
$4 + 6 \times 2 = 4 + 12 = 16$
Specific behaviours
✓ correct value

- (b) Use the formula $y = (x + 1)(x - 2)$ to calculate y when $x = 1.5$.

(2 marks)

Solution
$y = (2.5)(-0.5)$ $= -1.25$
Specific behaviours
✓ evaluates terms in brackets ✓ correct value

$$\begin{aligned}
 y &= (x+1)(x-2) \\
 y &= x^2 - 2x + x - 2 \\
 y &= 1.5^2 - 2(1.5) + 1.5 - 2 \\
 y &= 2.25 - 3 + 1.5 - 2 \\
 y &= 3.75 - 5 \\
 y &= -1.25
 \end{aligned}$$

[Ans only < correct ✓✓
Incorrect xx]

- (c) Use the formula $s = \frac{1}{2}at^2 - ut$ to calculate s when $a = 5.6$, $t = 10$ and $u = 12.5$. (2 marks)

Solution
$s = \frac{1}{2} \times 5.6 \times 10^2 - 12.5 \times 10$ $= 2.8 \times 100 - 125$ $= 280 - 125$ $= 155$
Specific behaviours
✓ substitutes and starts to simplify ✓ correct value

Question 6

(8 marks)

Consider the following matrices:

$$P = \begin{bmatrix} 4 & -5 \\ -3 & 2 \end{bmatrix} \quad Q = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \quad R = \begin{bmatrix} 2 & 0 \\ -3 & 1 \end{bmatrix} \quad S = \begin{bmatrix} 5 & -2 \end{bmatrix} \quad T = \begin{bmatrix} 3 \\ -2 \end{bmatrix} \quad U = \begin{bmatrix} -4 \\ 4 \end{bmatrix}.$$

(a) Calculate, where possible, the following. If not possible, give a reason why.

(i) $U - T$.

(1 mark)

Solution
$\begin{bmatrix} -4 \\ 4 \end{bmatrix} - \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -7 \\ 6 \end{bmatrix}$
Specific behaviours
✓ calculates result

(ii) $Q \times S$.

(1 mark)

Solution
Not possible, as number of columns in Q (2) is not the same as the number of rows in S (1).
Specific behaviours
✓ explanation using matrix dimensions

(iii) $2P + 3R$.

(2 marks)

Solution
$2 \begin{bmatrix} 4 & -5 \\ -3 & 2 \end{bmatrix} + 3 \begin{bmatrix} 2 & 0 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 8 & -10 \\ -6 & 4 \end{bmatrix} + \begin{bmatrix} 6 & 0 \\ -9 & 3 \end{bmatrix} = \begin{bmatrix} 14 & -10 \\ -15 & 7 \end{bmatrix}$
Specific behaviours
✓ calculates multiples ✓ calculates difference

↪ f/t if trying to add same size matrices

(iv) $R \times P$.

(2 marks)

Solution
$\begin{bmatrix} 2 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 4 & -5 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 8 & -10 \\ -15 & 17 \end{bmatrix}$
Specific behaviours
✓ at least two elements correct ✓ correct result

(b) Calculate R^2 .

(2 marks)

Solution
$R^2 = \begin{bmatrix} 4 & 0 \\ -9 & 1 \end{bmatrix}$
Specific behaviours
✓ calculates all four entries correctly ✓ calculates 2 or 3 entries correctly

Handwritten calculation for R^2 :

$$R \times R = \begin{bmatrix} 2 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ -9 & 1 \end{bmatrix}$$

↑ R^2

Question 7

(10 marks)

One weekend, a company operated three boats to run river trips.

On Saturday, boats P , Q and R carried 52, 66 and 45 adults, and 28, 42 and 22 children, respectively.

- (a) Represent this information in a 2×3 matrix X , with adults in the top row, and boats in alphabetical order. (1 mark)

Solution	
$X = \begin{bmatrix} 52 & 66 & 45 \\ 28 & 42 & 22 \end{bmatrix}$	
Specific behaviours	
✓ correct matrix	

$$X = \begin{matrix} & \text{BOATS} \\ & P & Q & R \\ \text{Adults} & 52 & 66 & 45 \\ \text{Children} & 28 & 42 & 22 \end{matrix}$$

On Sunday, a total of 97 adults and 88 children took trips. Boats Q and R carried an equal number of children and boat R carried 30 adults. Compared to Saturday, boat P took 19 fewer adults and 4 more children.

- (b) Represent the Sunday passenger numbers in matrix Y , using the same row and column labels as matrix X . (3 marks)

Solution	
Boat P : $52 - 19 = 33$ adults and $28 + 4 = 32$ children.	
Adults in boat Q : $97 - 33 - 30 = 34$.	
Split remaining $88 - 32 = 56$ children, $56 \div 2 = 28$ in boats Q and R .	
$Y = \begin{bmatrix} 33 & 34 & 30 \\ 32 & 28 & 28 \end{bmatrix}$	$\begin{matrix} P & Q & R \\ \text{ADULTS} \\ \text{CHILDREN} \end{matrix}$
Specific behaviours	
✓ column P ✓ column Q ✓ column R	

SUNDAY

$$Y = \begin{matrix} & P & Q & R \\ \text{ADULTS} & 33 & 34 & 30 \\ \text{CHILDREN} & 32 & x & x \end{matrix} \quad \begin{matrix} \text{TOTAL} = 97 \\ \text{TOTAL} = 88 \end{matrix}$$

$$\begin{aligned} 2x &= 88 - 32 \\ 2x &= 56 \\ x &= 28 \end{aligned}$$

- (c) Calculate matrix T , where $T = X + Y$ and explain what information it represents. (2 marks)

Solution	
$T = \begin{bmatrix} 52 & 66 & 45 \\ 28 & 42 & 22 \end{bmatrix} + \begin{bmatrix} 33 & 34 & 30 \\ 32 & 28 & 28 \end{bmatrix} = \begin{bmatrix} 85 & 100 & 75 \\ 60 & 70 & 50 \end{bmatrix}$ T shows the number of adults and children carried by boats P, Q and R over the weekend. Accept "in total"	
Specific behaviours	
✓ correct matrix T ✓ correct explanation	

*f/t from (a) and (b)
if (a) and (b)
show same size
matrices.*

The price per trip was \$10 per adult and \$5 per child.

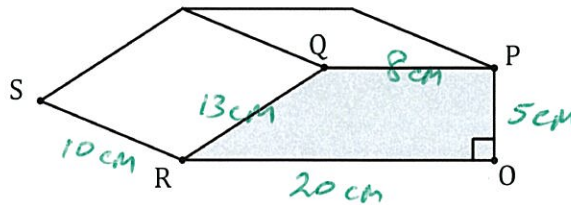
- (d) Represent the price information in matrix C that can be meaningfully multiplied by matrix T . Then calculate CT and explain what information it represents. (4 marks)

Solution	
$C = \begin{bmatrix} 10 & 5 \end{bmatrix}$ $CT = \begin{bmatrix} 10 & 5 \end{bmatrix} \begin{bmatrix} 85 & 100 & 75 \\ 60 & 70 & 50 \end{bmatrix} = \begin{bmatrix} 1150 & 1350 & 1000 \end{bmatrix}$ Matrix CT shows the total passenger fares collected over the weekend by each of the boats P, Q and R.	
Specific behaviours	
✓ correct matrix C (explicit or as shown in working) ✓ premultiplies a 1×2 matrix with a 2×3 matrix ✓ correct product ✓ correct explanation	

f/t

Question 8 *Penalty of -1 if no/wrong units for Q8 once or more.* (7 marks)

The diagram below, not drawn to scale, shows a prism with trapezoidal cross-section $OPQR$, where $OP = 5$ cm, $PQ = 8$ cm, $QR = 13$ cm, $OR = 20$ cm and $RS = 10$ cm.



- (a) Calculate the area of the cross-section $OPQR$. (2 marks)

Solution
$A = 8 \times 5 + \frac{1}{2} \times 12 \times 5$ $= 40 + 30 = 70 \text{ cm}^2$ <i>(or $A = (20+8) \div 2 \times 5 = 70$)</i>
Specific behaviours
✓ indicates appropriate method ✓ calculates area

*Ans only correct ✓
incorrect xx*

- (b) Calculate the volume of the prism. (2 marks)

Solution
$V = 70 \times 10$ $= 700 \text{ cm}^3$
Specific behaviours
✓ correctly uses answer from (a) ✓ calculates volume

f/t

- (c) Determine the total surface area of the prism. (3 marks)

Solution
Top, bottom, back and sloping face: $A = 10 \times (5 + 8 + 13 + 20)$ $= 10 \times 46 = 460$
Two ends: $A = 2 \times 70 = 140$
Total: $A = 460 + 140 = 600 \text{ cm}^2$
Specific behaviours
✓ indicates appropriate method for four rectangles ✓ includes both trapezoidal faces ✓ calculates total surface area

*Horizontal top $A = 8 \times 10 = 80$
Horizontal bottom $A = 20 \times 10 = 200$
Left sloping rect $A = 13 \times 10 = 130$
Right vertical rect $A = 5 \times 10 = 50$
Front trapezium $A = \frac{(20+8)}{2} \times 5 = 70$
Back trapezium $A = 70$*

f/t if 6 faces

T.S.A. = 600 cm²



Rossmoyne Senior High School

Semester One Examination, 2021

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One: Calculator-free

WA student number: In figures

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In words

Circle your teacher's name: Barrymore Fletcher Fraser-Jones Ng Pisano Younger

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

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- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
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Markers use only		
Question	Maximum	Mark
1	5	
2	5	
3	5	
4	7	
5	5	
6	8	
7	10	
8	7	
S1 Total	52	
S1 Wt (×0.6731)	35%	
S2 Wt	65%	
Total	100%	

Section One: Calculator-free

35% (52 Marks)

This section has **eight** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(5 marks)

- (a) Describe the type (using row, column, square, zero or identity) of each of the following matrices:

(i) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. (1 mark)

(ii) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. (1 mark)

(iii) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. (1 mark)

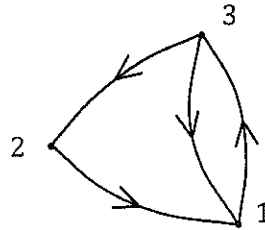
- (b) State the size of the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ in the form $m \times n$. (1 mark)

- (c) Write the 4×4 identity matrix I . (1 mark)

Question 2

(5 marks)

Three people who share a computer network have each forgotten their own password but can remember at least one other person's password. In the following diagram, the directed lines indicate that the person at the start of the line remembers the password of the person at the end of the line. For example, person 2 remembers the password of person 1.



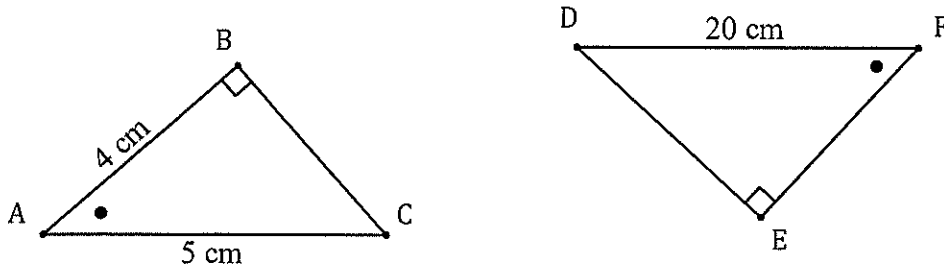
- (a) Create a square matrix M where the entry $M_{i,j}$ is 0 if person i does not remember the password of person j , and is 1 if they do. In your matrix list the people in numerical order, that is 1,2,3. (3 marks)

- (b) Determine M^2 . (2 marks)

Question 3

(5 marks)

The diagram below, not drawn to scale, shows two similar right triangles.



- (a) The larger triangle is the image and the smaller triangle is the original object. Calculate the scale factor for the larger triangle relative to the smaller. (1 mark)
- (b) Determine the length of side BC , the length of side DE and the length of side EF . State clearly which is which. (3 marks)
- (c) Calculate how many times greater the area of the large triangle is compared to the area of the small triangle. (1 mark)

Question 4**(7 marks)**

A young person has drawn up a budget for their weekly income of \$800. Some of their income is allocated to rent, living expenses, loan repayments, phone, and the rest to savings.

	A	B
1	Weekly budget	
2	Rent	\$235.00
3	Debt repayments	
4	Living expenses	\$290.00
5	Phone	\$65.00
6	Savings	
7	Total	\$800.00

The young person allocates 15% of their weekly income to debt repayments.

- (a) Determine the budget allocation for debt repayments. **(2 marks)**
- (b) The young person has been saving for a holiday and needs to save another \$720 to make the booking. If the above budget is maintained, determine the number of weeks will it take them to save this amount. **(3 marks)**
- (c) If the budget allocations for rent and phone increase by \$7 and \$3 respectively, what effect will this have on your answer to part (b)? **(2 marks)**

Question 5

(5 marks)

- (a) Calculate the value of $p + q \times r$ when $p = 4$, $q = 6$ and $r = 2$.

(1 mark)

- (b) By substituting for the given value of x , use the formula $y = (x + 1)(x - 2)$ to calculate y when $x = 1.5$.

(2 marks)

- (c) Use the formula $s = \frac{1}{2}at^2 - ut$ to calculate s when $a = 5.6$, $t = 10$ and $u = 12.5$. (2 marks)

Question 6**(8 marks)**

Consider the following matrices:

$$P = \begin{bmatrix} 4 & -5 \\ -3 & 2 \end{bmatrix} \quad Q = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \quad R = \begin{bmatrix} 2 & 0 \\ -3 & 1 \end{bmatrix} \quad S = [5 \quad -2] \quad T = \begin{bmatrix} 3 \\ -2 \end{bmatrix} \quad U = \begin{bmatrix} -4 \\ 4 \end{bmatrix}.$$

(a) Calculate, where possible, the following. If not possible, give a reason why.

(i) $U - T.$ (1 mark)

(ii) $Q \times S.$ (1 mark)

(iii) $2P + 3R.$ (2 marks)

(iv) $R \times P.$ (2 marks)

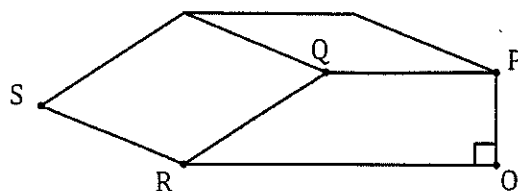
(b) Calculate $R^2.$ (2 marks)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 7

(7 marks)

The diagram below, not drawn to scale, shows a prism with trapezoidal cross-section $OPQR$, where $OP = 5$ cm, $PQ = 8$ cm, $QR = 13$ cm, $OR = 20$ cm and $RS = 10$ cm.



- (a) Calculate the area of the cross-section $OPQR$. (2 marks)

- (b) Calculate the volume of the prism. (2 marks)

- (c) Determine the total surface area of the prism. (3 marks)

Question 8**(10 marks)**

One weekend, a company operated three boats to run river trips.

On Saturday, boats P , Q and R carried 52, 66 and 45 adults, and 28, 42 and 22 children, respectively.

- (a) Represent this information in a 2×3 matrix X , with adults in the top row, and boats in alphabetical order. (1 mark)

On Sunday, a total of 97 adults and 88 children took trips. Boats Q and R carried an equal number of children and boat R carried 30 adults. Compared to Saturday, boat P took 19 fewer adults and 4 more children.

- (b) Represent the Sunday passenger numbers in matrix Y , using the same row and column labels as matrix X . (3 marks)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

- (c) Calculate matrix T , where $T = X + Y$ and explain what information it represents. (2 marks)

The price per trip was \$10 per adult and \$5 per child.

- (d) Represent the price information in matrix C that can be meaningfully multiplied by matrix T . Then calculate CT and explain what information it represents. (4 marks)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF